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Parth Soni

Senior Data & AI Engineer

AWS Community Builder | Content Creator

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Technology Development Professional with 8+ years of experience in **cloud-agnostic** and **platform-agnostic Data and AI** solution design across **AWS, Azure, and hybrid** cloud environments. Proven expertise in **Generative AI, Data Engineering, Streaming Systems, Machine Learning, and Modern Data Architectures**. Strong hands-on skills in **SQL, Python, Java, C#, and core Data Structures and Algorithms**. Adept at developing **code-agnostic**, scalable systems for **Data Warehousing, Analytics, and AI/ML workloads**, with a strong focus on adaptability, optimization, and delivering enterprise-grade, **AI-first solutions**.

SKILLS

Languages & Scripting	Python, PySpark, Scala, Java, SQL, C#, Bash/Shell
Big Data & Streaming	Kafka, Flink, PyFlink, Debezium, CDC, Spark, Hadoop, Delta Lake, Iceberg
Data Engineering & ELT	ELT Pipelines, Real-time Streaming, Distributed Systems, Data Warehousing, Data Modeling, dbt, Query Optimization
ML & GenAI	MLflow, SageMaker, Azure ML, LLMs, RAG, LangChain, Pinecone, PGVector
Cloud & DevOps	AWS (S3, EMR, Glue, Lambda, DynamoDB, Redshift, ECS/Fargate), Azure (Synapse, ADF), Docker, Kubernetes, Argo CD, Terraform, CI/CD, Jenkins, GitHub Actions, IAM, VPC
Workflow Orchestration	Apache Airflow, AWS Step Functions
Analytics & Activation	Funnel.io, Adobe Analytics, Event-driven Pipelines, Attribution, Personalization
Leadership & Collaboration	Architecture Design, Design Reviews, Mentoring, Cross-team Collaboration

TECHNICAL EXPERIENCE

Senior Data Engineer **June 2022 — Present**
Planet Payment *Bengaluru, Karnataka*

- Designed and developed **large-scale, real-time, fault-tolerant distributed data systems** using **CDC, Apache Kafka, Apache Flink/PyFlink, Debezium, Avro/Protobuf, and KRaft**, processing **billions of payment events daily** with low latency and strict SLAs.
- Led platform architecture and POCs using **Terraform (IaC)** and **GitHub-based CI/CD**, automating deployments via shell scripting; built custom **Docker images** and operated **Flink job clusters on Kubernetes** with **Argo CD** for GitOps-based rollouts.
- Orchestrated end-to-end ELT workflows using **Apache Airflow**, building DAGs with reusable data adapters to ingest data from **S3 and ERP APIs** into **Redshift raw layers**, followed by transformations into unified and analytics-ready warehouse layers.
- Automated **dbt** model execution via Airflow to generate curated and aggregated datasets, enabling reliable and repeatable business reporting and insights.
- Deployed and operated **Apache Airflow on AWS ECS Fargate**, integrating **Jenkins-based CI/CD pipelines** from GitHub to build, version, and deploy DAGs and platform changes.
- Incorporated **GenAI-assisted development and analytics** by leveraging **LLMs and RAG-based retrieval** to accelerate pipeline development, enable intelligent data discovery, and support AI-powered self-service insights on curated datasets.
- Collaborated closely with **Data Science, Product, and Engineering teams** to deliver **data- and AI-driven solutions**, including **feature engineering, real-time inference data flows, and RAG-based retrieval systems**, supporting personalization, fraud detection, and revenue optimization.
- Built and optimized **batch and streaming data pipelines** using **Python, PySpark, Spark (Scala), and dbt**, processing multi-terabyte datasets while reducing runtime and compute costs by **30–50%**.
- Architected **event-driven data products** using **Kafka Streams** with **DynamoDB/DocumentDB** as event stores, ensuring high availability, fault tolerance, and horizontal scalability.
- Designed and implemented a **Medallion Architecture** (Bronze, Silver, Gold) to standardize data ingestion, transformation, and analytics layers, with **Snowflake** serving as the Gold data warehouse for curated, business-ready datasets.
- Designed and deployed **metadata-driven pipelines** on **AWS EMR, Glue, Step Functions, Lambda, and S3** with built-in data quality checks, validation rules, and SLA monitoring.
- Built **analytical datasets and data cubes** using **Redshift, Hive, Parquet, ORC, and Delta Lake**, simplifying access to large and complex datasets for downstream analytics consumers.
- Implemented scalable **NoSQL ingestion and querying patterns** using **DynamoDB** with optimized partition keys and GSIs for high-throughput workloads.
- Designed secure, production-grade architectures using **IAM roles and policies, VPC endpoints, encryption, and least-privilege access controls**.
- Led **architecture discussions, design reviews, and code reviews**, mentoring engineers on **distributed systems, data engineering best practices, and AWS-native design patterns**.

Senior Application Engineer **April 2020 — June 2022**
Thomson Reuters *Hyderabad, Telangana*

- Designed and implemented **scalable data processing and ETL pipelines** using **Apache Spark, Hadoop, and Kafka** to support enterprise-scale analytics products.

- Built and maintained a centralized **data warehouse** using **star and snowflake schemas**, enabling efficient analytics and reducing manual reporting effort by **75%**.
- Developed **SQL-based transformation and reporting layers**, applying **query tuning, indexing strategies, and performance optimization** techniques.
- Automated ingestion and transformation workflows using **SSIS and Spark**, significantly improving data freshness and reliability.
- Reduced end-to-end data ingestion latency by **80%** through performance-driven redesign of pipelines and optimized data modeling.
- Led **cloud migration initiatives** from on-prem systems to **Azure (Synapse, Azure SQL, ADF, ADLS)** and **AWS (S3, EMR, Glue, Lambda, Kinesis)**, enabling scalable and cloud-native data platforms.

Software Development Engineer

Integration Point – Thomson Reuters

April 2018 – March 2020

Vadodara, Gujarat

- Developed **backend services and data integration components** using **C#/.NET**, supporting enterprise-grade transactional and analytical systems.
- Built **event-driven data synchronization pipelines** integrating Salesforce, **Azure Data Factory**, APIs, and NoSQL databases.
- Created **SQL-based monitoring and alerting solutions** to improve data pipeline observability and incident response.
- Partnered with security teams as an **Information Security Ambassador**, contributing to secure design reviews and compliance initiatives.

Software Development Intern

L&T Hydrocarbon Engineering Ltd

June 2017 – July 2017

Vadodara, Gujarat

- Contributed to enterprise applications using **ASP.Net, JavaScript, T-SQL, and Azure**, supporting internal business workflows.

EDUCATION

Bachelor of Engineering – Information Technology

Gujarat Technological University

2014 – 2018

CGPA – 8.73

Senior Secondary Education - Science

Kendriya Vidyalaya, CBSE

2012 – 2014

CGPA – 8.20

CERTIFICATIONS

AWS Certified Solutions Architect – Professional

2025 – 2028

AWS Certified Machine Learning – Specialty

2025 – 2028

AWS Certified Cloud Practitioner

Issued 2022

Microsoft Certified: Azure Data Engineer Associate

2021 – 2025

Microsoft Certified: Azure AI Fundamentals & Data Fundamentals

Issued 2021

Scrum Fundamentals Certified (SFC)

Issued 2020

PUBLIC SPEAKING

Practical Insights on Building Large-Scale Analytics Architectures

Apache Doris Summit 2025

<https://tinyurl.com/ApacheDorisSummitYT>

Showcasing high-performing data platform architecture(s) with Apache Doris and AWS

- AI-native real-time feature store: 70% cheaper than DynamoDB/Redis combo
- AI-powered intelligent log analytics
- Lakehouse intelligence layer on AWS
- Data platform with intelligent archival

Migrating from Snowflake to Apache Doris: Real-World Case Study

Apache Doris Webinar

<https://tinyurl.com/SFtoAWSDoris>

- Led the migration from Snowflake to Apache Doris, gaining deep hands-on experience and sharing insights to enable and upskill big data teams across the world.

Planet | TechTalks - Azure & AWS Data Engineering Services

Planet Tech Summit

<http://tinyurl.com/ParthAWS>

- Mentorship and knowledge sharing to build and grow cloud community.

<http://tinyurl.com/ParthAzure>

AWS Community Builder

AWS Events & Meetups

- AWS Community Builder recognized for delivering impactful talks on cloud-native AI, data platforms, and GenAI applications.